

CS & DA

Probability and Statistics

Permutations and Combinations

Lecture No. 01



By- Dr. Puneet Sharma Sir

Topics to be Covered



Topic

BASICS of P & C

(& Counting Principle)

GATE

RC → 2 hrs (1 class)

- PEC
- Prob
- Stats 1
- Stats 2

⇒ 62 hrs (31 classes)

Mon to Fri: → 8:30-10:30 PM

	CS/IT	DS/AI
	EM: 7-8M	EM: 36M
Linear-1	✓	✓
Linear-2	X	✓
Calculus	✓	✓
Prob & Stats-1	✓	✓
" " "-2	X	✓

(1) Live Class & Class Notes.

(2) DPP → Text Sol
→ Video Sol

(3) W.Test (Sunday) → 20 Marks ← 6Q → 1M
← 7Q → 2M

(4) P.Y.Q → PW APP (Book) → Engg Maths

(5) Tel Id: drpuneet&ir pw

Startegy : Step ① Live class

" ② Self Notes

" ③ Practice again of class Quest.

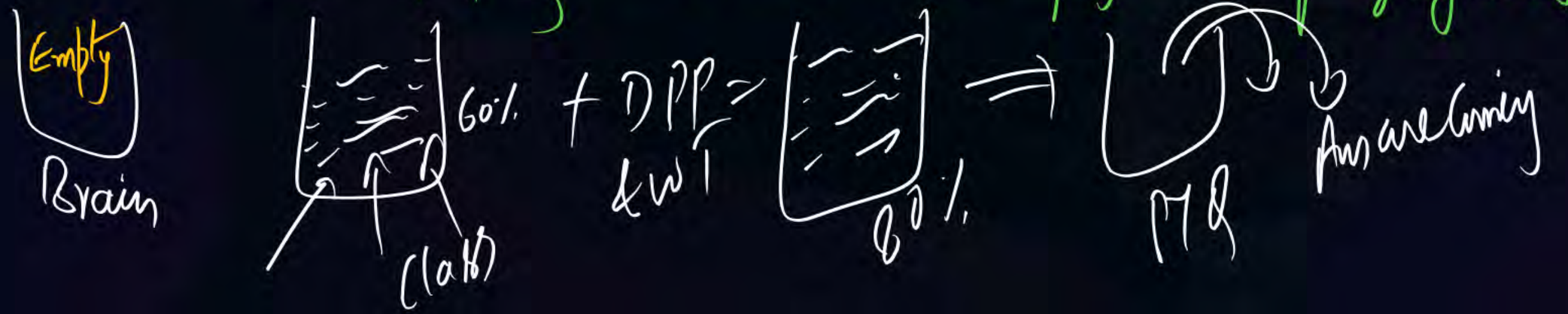
" ④ Solve DPP Questi — 60 to 70% Quest

" ⑤ W. Test

No Need to judge

" ⑥ PYQ — Now in this step you can judge yourself

80% PYQ



Perm & Combination & (Counting Principle)

Fundamental Principle of Addition → if we have to perform only one of the job at a time out of n jobs then use this principle. Keywords: "Either or / only one / Anyone"

Fundamental Principle of Multiplication → if we have to perform all the jobs at a time (out of n jobs) then use this Principle

Keywords: "AND / Both / All"

Q. there are 10 Boys & 8 Girls in a Class then in how many ways we can select

① Either a Boy or a Girls = ? $= 10 \text{ ways} + 8 \text{ ways} = 18 \text{ ways}$

$(B_1, B_2, B_3 \ldots\ldots, B_{10})$ or $(G_1, G_2, G_3 \ldots G_8)$

② A Boy & a Girl = ? $= 10 \text{ ways} \times 8 \text{ ways} = 80 \text{ ways}$

(B_1G_1) (B_2G_1) (B_3G_1) $\ldots\ldots$ $(B_{10}G_1)$
(B_1G_2) (B_2G_2) (B_3G_2) $(B_{10}G_2)$
(B_1G_3)
$\vdots$ $\vdots$ $\vdots$ $\vdots$
(B_1G_8) (B_2G_8) (B_3G_8) $\ldots\ldots$ $(B_{10}G_8)$
8 ways 8 ways 8 ways 8 ways

Qs There are (3) students appearing for Maths scholarship test, (4) for Physics & (5) are appearing for chemistry scholarship test then in how many ways these 8 scholarships can be awarded ?

Sol: Total ways of awarding 8 scholarships = 3 ways × 4 ways × 5 ways = 60 ways
(we have to award all 3 scholarships) (Maths) (Physics) (Chemistry)

Q: There are two friends and they have (5) jeans & (4) shirts in their Almirah (& they are sharing a room in Hostel) then in how many ways they can dress up for party?

Sol: Total ways to dress up $= \left(\frac{5}{J} \times \frac{4}{S}\right) \times \left(\frac{4}{J} \times \frac{3}{S}\right) = 240$ ways

f_1 f_2

$= (5 \text{ ways} \times 4 \text{ ways}) \times (4 \text{ ways} \times 3 \text{ ways}) = 240$ ways

J S J S

$\therefore$ Both the friends are going to attend f_1 Party & Each f has to wear J as well as S

f_2

Q. How many 3 digit Number can be formed using 1, 3, 5, 7, 9 if

① Repetition of digits are allowed (RA) = ? $= \underset{P_1}{5 \text{ ways}} \times \underset{P_2}{5 \text{ ways}} \times \underset{P_3}{5 \text{ ways}} = 125 \text{ ways}$

② " " " are not allowed (RNA) = ? $= \underset{P_1}{5 \text{ ways}} \times \underset{P_2}{4 \text{ ways}} \times \underset{P_3}{3 \text{ ways}} = 60 \text{ ways}.$

(∵ we have to fill (all) 3 places at a time (that's why we have Multiplied))

$= {}^5C_3 \times 3! = {}^5P_3 = 60$

Q. How many (3) letter words (with or w/o meaning) can be formed using Vowels

a, e, i, o, u

if (1) RA = ? = $\underset{P_1}{5 \text{ ways}} \times \underset{P_2}{5 \text{ ways}} \times \underset{P_3}{5 \text{ ways}}$ = 125 words

(2) RNA = ? = $\underset{P_1}{5 \text{ ways}} \times \underset{P_2}{4 \text{ ways}} \times \underset{P_3}{3 \text{ ways}}$ = 60 words = ${}^5C_3 \times 3! = {}^5P_3$

(*) Reason of Multiplication is same as that of previous Ques

(*) Here we have to fill places

Q. There are 4 letters and 5 letter Boxes then in how many ways we can post these letters ?

Wild Guesss: 4×5 ✗, $4+5$ ✗, ${}^5P_4$ ✗, ${}^5C_4$ ✗, $4!$ ✗, $5!$ ✗, 5^4 ✓, 4^5 ✗

Total ways of posting letters = 5 ways × 5 ways × 5 ways × 5 ways $= 5^4 = 625$ ways

(RA)

L_1 L_2 L_3 L_4

∵ we have to post (all) the letters that is why we will use Multi Rule

WRONG APP :- Total ways of posting letters = $\underset{B_1}{4}\times\underset{B_2}{4}\times\underset{B_3}{4}\times\underset{B_4}{4}\times\underset{B_5}{4} = (4^5)$ ✗

Here we are moving letter Boxes, which is not practically possible.

Q. there are 3 Rings & 4 fingers then in how many ways we can worn these Rings

if ① there is no Restriction =? (RA)

$= \underset{R_1}{4\text{ ways}} \times \underset{R_2}{4\text{ ways}} \times \underset{R_3}{4\text{ ways}} = 4^3 = 64\text{ ways}$

WRONG APP.:- Total ways $= \underset{f_1}{3} \times \underset{f_2}{3} \times \underset{f_3}{3} \times \underset{f_4}{3} = (3^4) \times$ (4 fingers) $\rightarrow R_1\ R_2\ R_3$

it is wrong bcoz in this Method we are cracking fingers

② with at Most one Ring in a single finger =? (RNA)



$= \underset{R_1}{4} \times \underset{R_2}{3} \times \underset{R_3}{2} = 24\text{ ways}$

$= {}^4(3 \times 3!) = {}^4P_3$

Q: There are 8 chairs then in how many ways 5 persons can be seated?

Sol: Total Seating Arrangements = $\underset{P_1}{8 \text{ ways}} \times \underset{P_2}{7 \text{ ways}} \times \underset{P_3}{6 \text{ ways}} \times \underset{P_4}{5 \text{ ways}} \times \underset{P_5}{4 \text{ ways}}$

(RNA)

$= 8 \times 7 \times 6 \times 5 \times 4 = {}^{8}C_5 \times 5! = {}^{8}P_5$

(i) If No. of chairs = (8) & No. of persons = (8) then T.S.A = ? $= 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

(RNA)

$= {}^{8}P_8 = 8!$

(ii) If No. of Chairs = 5 & No. of persons = 8 then T.S.A = ? = Senseless Question

Conclusion:- Let n = No. of places, r = No. of persons

① if $n > r$ & RNA then Multi Rule = Perm Rule

② if $n = r$ & RNA then Multi Rule = Perm. Rule = Factorial Rule

③ if RA then only use Multi Rule.

i.e the concept of ${}^{n}C_r$, ${}^{n}P_r$ & $r!$ exist only when R.N.A

(HW8) 5 persons entered in a lift at Ground floor in 8 floor house.
Then in how many ways they can leave the lift

① At any floor = ? (RA)
$= \underset{P_1}{7} \times \underset{P_2}{7} \times \underset{P_3}{7} \times \underset{P_4}{7} \times \underset{P_5}{7} = 7^5$ ways

② At different floors = ? (RNA)
$= \underset{P_1}{7} \times \underset{P_2}{6} \times \underset{P_3}{5} \times \underset{P_4}{4} \times \underset{P_5}{3} = {}^{7}P_5$ ways



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THANK - YOU